

Lem Invarianza Derivada Hiperbolica Aut Disco Unidad

Lema 1 (Invarianza derivada hiperbólica por $\text{Aut}(\mathbb{D})$).

Sea $T \in \text{Aut}(\mathbb{D})$ y $f \in \mathcal{H}(\mathbb{D})$. Entonces, $(f \circ T)^{\natural} = f^{\natural} \circ T$.

Demostración: Como $T \in \text{Aut}(\mathbb{D})$, por el teorema de Schwarz-Pick, $|T'(z)| = \frac{1-|T(z)|^2}{1-|z|^2}$.
Entonces, al aplicar la definición de derivada hiperbólica, tenemos que

$$\begin{aligned} \forall z \in \mathbb{D} : (f \circ T)^{\natural}(z) &= \frac{1-|z|^2}{2} |(f \circ T)'(z)| = \frac{1-|z|^2}{2} |f'(T(z))| \cdot |T'(z)| \\ &= \frac{\cancel{1-|z|^2}}{2} |f'(T(z))| \cdot \frac{1-|T(z)|^2}{\cancel{1-|z|^2}} = f^{\natural}(T(z)). \end{aligned}$$

■

Referenciado en

- Cor Sup Derivada Hiperbolica Aut Disco Unidad

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.